

SOP-07: Corrosive Inorganic and Organic Acids (PHS - Sulfuric Acid)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Rooms 209 and 213
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Sulfuric acid 90-100%	7664-93-9
Hydrochloric acid 10%-33% (concentrated)	7647-01-0
Hydrochloric acid solution, 0.2N (certified)	7647-01-0
Glacial acetic acid	64-19-7
pH 4.01 / 4.00 buffer solutions (dilute acid; low hazard, included for completeness)	mixture

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Corrosion (skin corrosion Cat 1, eye damage Cat 1, corrosive to metals); Health hazard (sulfuric acid, carcinogen Cat 1A as strong inorganic acid mist); Flame (glacial acetic acid, flammable Cat 3); Exclamation mark

- PARTICULARLY HAZARDOUS SUBSTANCE. Concentrated sulfuric acid is classified GHS Carcinogenicity Category 1A - strong inorganic acid mists containing sulfuric acid are an IARC Group 1 human carcinogen (laryngeal and lung cancer). Sections 9-11 of this SOP are mandatory and apply to all acids covered here.

- Stock chemicals: sulfuric acid 90-100% (500 mL), hydrochloric acid 10-33% (1 L), 0.2N hydrochloric acid (500 mL), glacial acetic acid (2.5 L), and commercial pH 4 calibration buffers. Intermediates and working solutions: dilute acid solutions prepared for pH adjustment of seawater and buffers, acid washes for glassware and trace-metal-clean labware, and 1X TAE buffer containing dilute acetic acid. Wastes: spent acid solutions, acid rinses, and neutralized aqueous waste where treatment is authorized.
- Routes of exposure: skin and eye contact (primary), inhalation of mist, vapor or aerosol, ingestion.
- How exposure might occur: pouring or pipetting concentrated acid, diluting acid, splash from an exothermic dilution, acid washing of glassware and plasticware, opening a warm or pressurized container, and contact with acid-wetted surfaces.
- Target organs: skin, eyes, respiratory tract, teeth (chronic acid mist exposure causes dental erosion), and gastrointestinal tract on ingestion. Concentrated sulfuric acid mist targets the larynx and lung.
- Signs and symptoms of exposure: immediate pain, redness and burns on skin contact; severe eye pain and possible permanent corneal damage or blindness; coughing, choking and chest pain on inhalation; hydrochloric acid vapor causes upper airway irritation at low concentrations. Acid burns from glacial acetic acid may appear mild at first and then blister.
- Physical and reactivity hazards: sulfuric acid is a Class 1 oxidizer and a Class 2 water-reactive material - dilution is strongly exothermic and can boil and spatter acid violently. ALWAYS ADD ACID TO WATER, NEVER WATER TO ACID. Sulfuric acid reacts violently with bases, with organic material (it chars paper, cloth and skin), and with metals to release flammable hydrogen. Hydrochloric acid attacks metals to release hydrogen and reacts with bleach to release chlorine gas, and with cyanide or guanidinium salts to release hydrogen cyanide. Glacial acetic acid is a Class II combustible liquid (flash point 39 degC) and freezes at 17 degC, so a bottle may solidify and crack.
- Expected by-products: hydrogen gas from acid-metal contact; chlorine from HCl plus bleach; hydrogen cyanide from HCl plus guanidinium-containing extraction waste - see SOP-05.
- GHS classification (from MyChem / SDS): Sulfuric acid - Carcinogenicity Category 1A; Corrosive to metals Category 1; Serious eye damage Category 1; Skin corrosion Category 1; STOT-single Category 3; Acute aquatic toxicity Category 3; Oxidizer Class 1; Water Reactive Class 2. Hydrochloric acid 10-33% - Corrosive to metals Category 1; Serious eye damage Category 1; Skin corrosion Category 1B; STOT-single Category 3. HCl 0.2N - Corrosive to metals Category 1; Skin corrosion Category 2; eye irritation Category 2. Glacial acetic acid - Flammable liquids Category 3; Serious eye damage Category 1; Skin corrosion Category 1A.
- Occupational exposure limits: sulfuric acid 1 mg/m³ (WISHA/OSHA PEL, 8-hr TWA; ACGIH TLV 0.2 mg/m³ thoracic fraction, A2); hydrogen chloride 5 ppm ceiling; acetic acid 10 ppm 8-hr TWA.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- All dispensing, dilution and transfer of concentrated sulfuric acid, concentrated hydrochloric acid, and glacial acetic acid is performed inside a certified chemical fume hood with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S. FTR 209 has the certified fume hood; acid work originating in FTR 213 is carried to FTR 209 in secondary containment for these operations, or performed in a certified hood in FTR 213 if one is available and current.
- Dilute acids (0.2N HCl, pH calibration buffers, 1X TAE) may be handled at the bench.
- Acids are stored and used away from metals, bases, oxidizable organics, bleach and guanidinium-containing materials. A corrosives storage cabinet with an acid-resistant liner and secondary containment trays is used in both rooms; sulfuric acid, an oxidizer, is kept physically separated from the organic acid (glacial acetic acid) within that cabinet.
- An acid spill kit containing a neutralizing absorbent (sodium bicarbonate or a commercial acid neutralizer with color indicator) is maintained in FTR 209 and FTR 213.

- The nearest eyewash and safety shower in each room are inspected and flushed weekly by lab staff, and their location is confirmed and the path kept clear before any concentrated acid work begins. Acid burns require immediate 15-minute flushing; a blocked eyewash is a stop-work condition.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Because these chemicals are corrosive and toxic by skin contact, additional protective clothing is REQUIRED where splashes or skin contact are foreseeable: a chemically resistant (PVC or neoprene) apron plus a face shield for any handling of concentrated acid, any dilution, and all spill cleanup. Disposable chemically resistant sleeves are worn for transfers of more than 100 mL.
- Acid-contaminated clothing is removed immediately, under the safety shower if the skin is involved, and is discarded rather than laundered if saturated with concentrated acid.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: for concentrated sulfuric acid, concentrated hydrochloric acid and glacial acetic acid, wear butyl rubber or neoprene gloves (e.g., Ansell AlphaTec Butyl 38-612, or North Neoprene) over an inner 4-6 mil nitrile glove. Thin nitrile alone is NOT adequate for concentrated acids.
- For dilute acids (0.2N HCl, pH buffers, TAE), 4-6 mil nitrile gloves are sufficient, changed on any contact.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Any glove contacted by concentrated acid is removed and replaced immediately; do not attempt to rinse and reuse it.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all work with concentrated acids, including small transfers. A face shield worn over goggles is REQUIRED for all dilution of concentrated acid, all transfers of more than 100 mL, and all spill cleanup. Safety glasses with side shields are acceptable for dilute acids and pH buffer use.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP when they are performed as written.
- Concentrated sulfuric acid, concentrated hydrochloric acid and glacial acetic acid must NEVER be used outside of a certified chemical fume hood. Dilute acids (0.2N HCl, pH buffers, TAE) may be used at the bench.

- No lab member is currently enrolled in the EH&S Respiratory Protection Program for these chemicals. If work is proposed that cannot be conducted in a fume hood, work must stop and the PI must contact EH&S.
- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.
- Where air-purifying respirators are appropriate, use a full-face respirator with acid gas cartridges as a backup to engineering controls. Use a full-face supplied air respirator if it is the sole means of protection.

Section 4 - Special Handling and Storage Requirements

- ALWAYS ADD ACID TO WATER, NEVER WATER TO ACID. Add concentrated acid slowly, in small increments, down the side of the vessel, with stirring, and with the vessel in an ice-water bath for sulfuric acid dilutions of more than 50 mL.
- Quantity limits: no more than 100 mL of concentrated acid is dispensed at one time; no more than 500 mL of concentrated acid is out of the corrosives cabinet at once. Operations requiring more than 250 mL of concentrated sulfuric acid require prior written PI approval.
- Use glass or acid-compatible fluoropolymer labware. Do not use metal spatulas, metal stir bars without a coating, or metal-containing equipment with acids.
- Never mix acid with bleach, with guanidinium-containing extraction waste (SOP-05), with sodium hydroxide (SOP-08) except during controlled, approved neutralization, or with oxidizer solutions.
- Never combine sulfuric acid with hydrogen peroxide - this makes piranha solution, which is not authorized in this laboratory under this SOP and requires a separate EH&S-reviewed process SOP.
- All working containers and dilutions must be labeled with the full chemical name, the concentration, the word 'CORROSIVE', the date and the preparer's initials. Sulfuric acid containers additionally carry 'CANCER HAZARD - ACID MIST'.
- Concentrated acid work is performed only when at least one other person is present in or immediately adjacent to the room. Unattended operations with open concentrated acid are prohibited.
- A 'CORROSIVE ACID IN USE' sign is posted on the fume hood sash while work is in progress; the sulfuric acid sign adds 'PARTICULARLY HAZARDOUS SUBSTANCE'.
- Storage: acids are stored below eye level in the acid/corrosives cabinet in FTR 209 and FTR 213, in acid-resistant secondary containment trays, with inorganic acids segregated from organic acid (glacial acetic acid) and from all bases, oxidizable organics, flammables, cyanides and metals. Glacial acetic acid is stored above 17 degC to prevent freezing and container cracking. Do not store acids under a sink or on the floor.
- Transport: bottles are carried in acid-resistant bottle carriers. Acids moved between FTR 209 and 213 travel in a closed secondary container by the most direct route.
- Best practices to minimize accidents: pre-stage everything inside the hood before opening a bottle; keep neutralizer within reach; never pipette by mouth; never return unused acid to the stock bottle.
- Clean the chemical fume hood upon completion of tasks with water followed by a dilute sodium bicarbonate solution and then clean water; wipe dry.
- Clean all contaminated surfaces with water and dilute sodium bicarbonate solution, rinse with clean water, and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.

- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with these acids must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- Small spill - less than 100 mL of concentrated acid contained INSIDE the operating fume hood, or less than 250 mL of dilute acid at the bench: may be handled by trained lab staff. Wearing goggles, face shield, butyl or neoprene gloves over nitrile, lab coat with apron and sleeves, apply acid-neutralizing absorbent from the spill kit around the perimeter first and then inward, wait for the indicator to show neutralization is complete, and scoop the residue into a screw-top container using a plastic scoop.
- Any concentrated acid spill greater than 100 mL, any spill outside the fume hood, any spill that reaches a metal surface, drain or electrical equipment, or any spill in which a person may have been exposed: DO NOT clean up. Evacuate the room, close the door, post a warning sign, prevent entry, and call 9-1-1 from a campus phone. Notify EH&S.
- Do NOT flush concentrated acid spills to the drain with water; adding water to concentrated sulfuric acid on a bench generates heat and spatter. Neutralize first.
- Liquid spills are the expected form. Solidified glacial acetic acid is allowed to warm and liquefy inside the hood before handling; do not chip it.
- PPE required for cleanup: chemical splash goggles AND face shield, butyl or neoprene gloves over 6-8 mil nitrile, buttoned lab coat with chemically resistant apron and disposable sleeves, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - ACID SPILL' on the door for any spill outside the hood; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after neutralizer removal, verify neutrality with pH paper, then clean the spill area thoroughly with detergent solution followed by clean water, and dry.
- Spill cleanup materials must be disposed of in the following manner: place neutralized residue and contaminated disposables in a double-bagged, screw-top container, label with a UW Hazardous Waste Label listing the acid and 'neutralized spill debris', and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure: use the nearest safety shower IMMEDIATELY for at least 15 minutes; stay under the shower and remove all contaminated clothing including shoes and jewelry; use a clean lab coat or spare clothing for cover-up. Do not apply neutralizers, creams or ointments to acid burns. Seek medical evaluation for any burn from concentrated acid.

- Eye exposure: use the eyewash for at least 15 minutes while holding the eyelids open, then go immediately to the Emergency Department. Concentrated acids cause irreversible eye damage; do not delay to look for someone to help.
- Inhalation exposure: move the person out of the contaminated area to fresh air; get medical help. Acid mist or hydrogen chloride inhalation can cause delayed pulmonary edema - medical evaluation is required even if the person feels better.
- Ingestion: do NOT induce vomiting. Rinse the mouth with water and give small sips of water if the person is conscious and able to swallow. Call 9-1-1 immediately.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent and concentration, dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused or expired concentrated acid; (b) spent dilute acid solutions and acid washes; (c) acid rinses from glassware; (d) acid-contaminated solid debris (absorbent, gloves, wipes); (e) neutralized aqueous waste where in-lab neutralization is authorized.
- Keep acid waste SEPARATE from base waste, from oxidizer waste, from halogenated and non-halogenated solvent waste, from bleach, and above all from guanidinium-containing extraction waste (SOP-05) - acidifying guanidinium waste generates hydrogen cyanide.
- In-lab treatment: dilute acid solutions of pH 2 or above that contain no other hazardous constituent may be neutralized to pH 6-9 with sodium bicarbonate or dilute sodium hydroxide and, if EH&S guidance for the specific solution permits, disposed of to the sanitary sewer with copious water. Neutralization is performed inside the fume hood, slowly, with stirring and pH paper verification, wearing full acid PPE. Concentrated sulfuric acid, concentrated hydrochloric acid, glacial acetic acid, and any acid containing metals, solvents or other hazardous constituents are NOT neutralized in the lab and are collected as hazardous waste.
- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible container (acid-resistant HDPE) with a securely closable screw-top lid, kept closed except when actively adding waste, in acid-resistant secondary containment. Do not use metal containers or metal caps.
- Never seal a container of acid waste that may still be reacting or evolving gas.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Be aware that many laboratory accidents happen from inadvertent disposal of incompatible wastes into the same waste container. Therefore, identify different waste streams as appropriate.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.

- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplus materials. Update the MyChem inventory when containers are emptied, received, transferred or surplus.

Section 7 - Protocol

- Procedure A - Dilution of concentrated acid. Inside the certified fume hood, wearing goggles, face shield, butyl over nitrile gloves, lab coat, apron and sleeves: place the required volume of deionized water in a heat-resistant glass or plastic vessel, set it in an ice-water bath for sulfuric acid, and add the concentrated acid slowly down the side of the vessel with continuous stirring. Allow the solution to cool to room temperature before transferring or capping. Label immediately.
- Procedure B - pH adjustment of buffers and seawater. Use dilute acid (0.2N HCl or 1M HCl prepared per Procedure A) added dropwise with stirring and continuous pH monitoring. This may be done at the bench with nitrile gloves and safety glasses.
- Procedure C - Acid washing of labware for trace metal work. Inside the fume hood, fill an acid-resistant bath with dilute (typically 10%) hydrochloric or nitric acid prepared per Procedure A, submerge labware, cover, and soak for the interval specified in the protocol. Remove with plastic forceps, rinse three times with deionized water into the acid rinse waste container, and dry in a clean area. Post a sign on the covered bath giving contents, owner and date.
- Procedure D - pH meter calibration. Use commercial pH 4.01, 7.00 and 10.01 buffers at the bench with nitrile gloves and safety glasses. Used buffer may be discarded to the drain with running water; these are dilute, non-hazardous commercial solutions. Verify with the SDS before drain disposal.
- Prohibited: adding water to concentrated acid; combining sulfuric acid with hydrogen peroxide (piranha); mixing acid with bleach or guanidinium waste; storing acid with bases, metals or flammables; use of concentrated acid outside a certified fume hood; use by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - acids are not administered to animals under this SOP. Dilute acid is used only for pH adjustment of experimental seawater; if seawater acidification exposes live animals, the exposure is conducted under the applicable IACUC-approved protocol and the acid handling still follows this SOP.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with the acids covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each acid, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.

- Concentrated sulfuric acid is a particularly hazardous substance. Written authorization from the Principal Investigator is required before any individual may work with concentrated sulfuric acid, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDSs; (4) documented instruction on acid dilution technique, acid-base incompatibility, and the location and use of the eyewash and safety shower.
- A worker must complete at least one supervised dilution (Procedure A), observed and signed off by the PI or a designated experienced lab member, before diluting concentrated acid independently for the first time. Two supervised runs are required before independent work with concentrated sulfuric acid.
- Any new process combining acids with other chemicals, and any use of more than 250 mL of concentrated sulfuric acid, requires prior written PI approval and consultation with EH&S.
- No medical examination is required for the work described because respirators are not used. A medical examination and fit testing would be required before any respirator use.

Section 10 - Decontamination

- Fume hood: wipe the work surface and inside of the sash with clean water, then dilute sodium bicarbonate solution, then clean water; verify neutrality with pH paper and dry.
- Equipment: rinse pipettes, cylinders and stir bars into the acid rinse waste container, then wash with detergent and water and rinse three times with deionized water. Inspect metal equipment near acid work for corrosion and replace as needed.
- Glassware: rinse three times inside the fume hood, collecting the first rinse as acid waste, then wash with detergent and water and rinse three times with deionized water.
- Controlled areas: there are no glove boxes, restricted-access hoods or perchloric/hot acid hoods in the Roberts Lab space. Perchloric acid work is not performed and is not authorized under this SOP.
- Cleaning solutions and materials: clean water, dilute (5-10%) sodium bicarbonate solution, pH paper, laboratory detergent solution, deionized water, lint-free wipes, and acid-neutralizing absorbent.

Section 11 - Designated Area

- DESIGNATED AREA: the certified chemical fume hood in FTR 209 is the designated area for all use, dilution, dispensing and waste transfer of concentrated sulfuric acid, concentrated hydrochloric acid and glacial acetic acid. If a certified fume hood in FTR 213 is used for this work, it is likewise a designated area and is posted accordingly.
- The designated area is posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - CONCENTRATED SULFURIC ACID - CORROSIVE - CANCER HAZARD (ACID MIST)' sign listing the PI, contact number, required PPE and the authorization requirement.
- Acid waste containers are kept within the designated area in acid-resistant secondary containment.
- Storage: the acid/corrosives cabinets in FTR 209 and FTR 213, below eye level, with secondary containment and inorganic/organic acid segregation; both cabinets are posted with corrosive hazard signage and the sulfuric acid position carries the PHS designation.
- Dilute acids (0.2N HCl, commercial pH buffers, 1X TAE) may be used at any laboratory bench in FTR 209 and FTR 213 with the PPE specified in Section 3.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

[illegible]